

The paper **"Localized nonlinear dissipative matter waves controlled by quantum fluctuations"** address the interesting problem of solitons in Bose Einstein condensates in an effort to shed light on both. This is a tempting reserach direction given the similarities between solitons and Bose Einstein condensates, both generally described by some form of the nonlinear schrödinger equation, the latter hodling generally hodling an additional term wichi is the influence of some external field.

In this study, the authors go beyond the well known mean field approximation and study the dynamics of the system up to the 4th order in an effort to better take into account the effects of the quantum fluctuations in the system.

1 Theory

The basic expression to address BEc is the following Gross-Pitaevskii equation

$$i\hbar \frac{\partial \psi}{\partial t} = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V_{\text{ext}}(x) + g |\psi|^2 \right] \psi, \quad (1)$$

where g is the interaction strength and V_{ext} is the external trapping potential, usually in an harmonic form. Let us note that the presence of $V_{\text{ext}}(x)$ is the main difference between the above Gross Pitaevskii equation and the nonlinear Schrödinger equation below used to describe solitons in general :

$$i\hbar \frac{\partial \psi}{\partial t} = \left[-\frac{1}{2} \frac{\partial^2}{\partial x^2} + g |\psi|^2 \right] \psi. \quad (2)$$

The harmonic trap in the Bose Einstein condensate (1) ensures the localization of the matter wave in finite region of space.

In this study however, the authors go beyond the mean field approximation expressed by the neglectation of the higher order term and go up he 4th order

$$i\hbar \frac{\partial \psi}{\partial t} = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V_{\text{ext}}(x) + g_2 |\psi|^2 + g_3 |\psi|^3 + g_4 |\psi|^4 + i \frac{\gamma}{2} \right] \psi, \quad (3)$$

The additional terms help modelize the nature of their specific system which takes into account :

- the quantum fluctuations $\sim |\psi|^3$
- the three body loss effect $\sim |\psi|^4$
- the feeding parameter $\sim \gamma$

The precise details which further charaterize their system surch as the correction to the external potential caused by the cheosen optical lattices are not explicited so we can keep it concise.

2 Method

The authors then proceed to solve (3) by mean of the variational approach, using a classical gaussian trial function

$$\psi(c, \tau) = A(\tau) \exp \left\{ -\frac{c^2}{2a^2(\tau)} + \frac{i}{2}b(\tau)c^2 + i\theta(\tau) \right\} \quad (4)$$

in the Lagrangian of the system.

Inserting the previous trial function in the Lagrangian of the system, they are left with the differential equations for the parameters A , a and b which are then solved and analyzed for stability, by the means of the Vakhitov-Kolokolov (VK) condition, that states that one of the condition for the stability of the soliton is $\partial N / \partial \mu < 0$ everywhere, where N is the number of particles and μ the chemical potential

3 Results

They apply their model to a sample of $N(0)$ atoms of ^7Li , with some specific parameters (see the paper for details). They then study the evolution of the free variational parameters $A(\tau)$, $a(\tau)$, and $b(\tau)$ for $N(0) = 800$ (Fig. 1), and $N(0) = 3500$ (Fig. 2).

In Fig. 3, they study the stability of $|\psi|^2$ as a function of space and time under their model for one soliton.

In Fig. 4, 5 and 6, they apply the same study as the previous one for a sample of $N(0) = 3500$ and different parameters and prove the stability of their model under said parameters.

In this study, the authors build a matter wave and show its stability under different sets of parameters $A(\tau)$, $a(\tau)$ and $b(\tau)$, as well as different values of the binary interaction g_2 (see Eq. (3)). Doing so, they show the possible existence of localized modes in a trapped BEC with three body recombination, atomic feeding and quantum fluctuation.

Moreover, they find that the initial Gaussian form from their trial function breaks into three trains of solitons which oscillate during propagation and shed light on the effect of the strength of the quantum fluctuations on the the period of oscillations and the spatial separation of localized matter wave.

This study will contribute in the general understanding of the dynamics of matter wave and will find its importance in BEC applications.

4 Conclusion and personal considerations

In this paper, Mboumba *et al.* study the stability of matter waves, in this case solitons, trapped into Bose-Einstein condensates. Their model takes into account quantum fluctuations, three body losses as well as atomic feeding. They design the Gross-Pitaevskii equation with the appropriate parameters for stability and demonstrate the viability of the system. From a methodological point of view, they make use of the variational method in order to find an appropriate test function, and derive from the Lagrangian, the necessary values for the stability of the system.

This study demonstrates the role of quantum fluctuations in the dynamics and allows to have an insight of the spatial behaviour of the system in space, during propagation. The main goal is to gain some insight that will later allow to control the solitons in BEC for applications. Nonetheless, the study is highly theoretical and the authors fail to stress, as far as that paper is concerned, the link between some current experimental data and their actual treatment. If not possible, it would have been very good if they at least compared their data with previous results not taking into account quantum fluctuations or three body losses, for example.